United States Patent	12385310
Kind Code	B2
Date of Patent	August 12, 2025
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Furniture hinge and piece of furniture with such a furniture hinge

Abstract

A furniture hinge includes a first hinge part for attachment to a furniture door or flap, and a second hinge part for attachment to a furniture body. A hinge assembly movably connects the first hinge part to the second hinge part. A damping element dampens an opening movement and/or a closing movement of the hinge assembly at least towards the end of the movement. The damping element includes a cylinder, a piston linearly movable in the cylinder, and a piston rod coupled to the piston and projecting from the cylinder. The cylinder of the damping element is fastened pivotably about a pivot axis directly or indirectly to one of the hinge parts at least during the opening movement and/or closing movement of the furniture hinge.

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Appl. No.:	**18/272478**
Filed (or PCT Filed):	**January 25, 2021**
PCT No.:	**PCT/TR2021/050058**
PCT Pub. No.:	**WO2022/159048**
PCT Pub. Date:	July 28, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20240068291 A1	Feb. 29, 2024

Publication Classification

Int. Cl.: **E05F1/08** (20060101); **E05F5/00** (20170101)

U.S. Cl.:

CPC **E05F5/006** (20130101); E05Y2900/20 (20130101)

Field of Classification Search

CPC: E05Y (2900/20); E05Y (2900/202); E05Y (2900/204); E05Y (2900/208); E05Y (2201/20); E05Y (2201/21); E05Y (2201/261); E05F (5/006); E05F (5/02); E05F (5/027); E05D (11/1021); E05D (11/1042); E05D (11/105); E05D (11/1064); E05D (7/04); E05D (7/0407); E05D (7/125); E05D (3/142); E05D (16/286); E05D (16/258)

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Background/Summary

BACKGROUND OF THE DISCLOSURE
Field of the Disclosure
(1) The present invention refers to a furniture hinge for fastening a furniture door or flap of a piece of furniture to a furniture body of the piece of furniture in an articulated manner. The furniture hinge comprises: a first hinge part for attachment to the furniture door or flap, and a second hinge part for attachment to the furniture body, a hinge assembly which movably connects the first hinge part to the second hinge part in order to realise an opening movement of the furniture hinge, in which the first hinge part moves away from the second hinge part, and an oppositely directed closing movement of the furniture hinge, at least one damping element for damping the opening movement and/or the closing movement of the furniture hinge at least towards the end of the movement, the damping element comprising a cylinder, a piston which is linearly movable therein relative to the cylinder, and a piston rod which is coupled to the piston and projects from the cylinder, and a mounting piece which is attached to the furniture body and to which the second hinge part is attached
(2) Furthermore, the invention refers to a piece of furniture comprising a furniture body and a furniture door or flap fastened to the furniture body in an articulated manner by means of at least one furniture hinge.
Description of the Prior Art
(3) Such furniture hinges and pieces of furniture with such furniture hinges are known in various embodiments from the prior art. In this context, reference is made by way of example to DE 101 59 140 A1, DE 20 2006 001 648 U1, EP 1 555 372 A1 and DE 10 2007 031 175 B3. Usually, the damping element is rigidly attached to or in the second hinge part. Actuation of the damping element is effected via the piston rod through the first hinge part, which is movable relative to the second hinge part, or through another component of the furniture hinge attached to the first hinge part. These known furniture hinges have the disadvantage that, during the opening movement and/or closing movement of the furniture hinge, transverse forces, i.e. forces directed transverse to the direction of movement of the piston rod in the cylinder of the damping element, are exerted on the piston rod. These transverse forces can weaken the damping element over time, for example by impairing the tightness of the piston in the cylinder or a bearing of the piston in the cylinder. The result can be leakage and/or a defect of the entire damping element. To overcome this disadvantage,

it is known to provide deflection gears between the first hinge part or the other component of the furniture hinge attached thereto on the one hand and the piston rod of the damping element on the other hand, which are intended to transform the non-linear movement of the first hinge part or the other component of the furniture hinge attached thereto into a actuating movement that is as straight as possible and which acts in the direction of movement of the piston rod. However, it is disadvantageous that the deflection gears require additional components, which increases material and assembly costs. In addition, the deflection gears are mechanically susceptible and require a relatively large installation space.

(4) Another furniture hinge is known from WO 2010/102 445 A1 in which the damping element is attached to the mounting piece so that it can pivot about a pivot axis in respect to the mounting piece. However, this requires a relatively large installation space and only properly works in the proposed manner only if the mounting piece is angled. For an adjustment of the furniture hinge or of the furniture door or flap in respect to the furniture body, the relative position of the second hinge part in respect to the mounting piece can be adjusted. In the case of the hinge described in this document, an adjustment of the relative position results in a variation of the position of an articulation point on the piston rod of the damping element, while the pivot axis through which the cylinder of the damping element is pivotably fastened to the mounting piece remains unchanged. This results in an undesired variation of the damping properties or damping characteristics of the damping element.

(5) Based on the described prior art, it is an object of the present invention to provide for a compact, inexpensive and durable furniture hinge with damping properties.

SUMMARY OF THE DISCLOSURE

(6) To solve this problem, a furniture hinge and a piece of furniture are proposed. In particular, starting from the furniture hinge of the type mentioned at the beginning, it is suggested that the cylinder of the damping element is fastened to one of the hinge parts or to a first component of the furniture hinge, which is rigidly connected to one of the hinge parts at least during the opening movement and/or closing movement of the furniture hinge, pivotably about a pivot axis.

(7) The first hinge part is, for example, a hinge cup, and the second hinge part is, for example, a hinge arm. The hinge cup is preferably placed in a recess on the inside of the furniture door or flap in a known manner and fastened thereto, for example by means of screws. The hinge arm is preferably attached to the mounting piece, which in turn is attached to the furniture body on the inside of the piece of furniture, e.g. by means of screws. The hinge arm can be attached to the mounting piece, e.g. by means of screws, a snap-in connection or the like. The attachment of the hinge arm to the mounting piece can be fixed, adjustable and/or releasable. For example, it is conceivable to fix the hinge arm, and with it those components of the furniture hinge attached to it and the furniture door or flap, in a desired position and/or orientation with respect to the mounting piece by means of screws. It would also be conceivable to use one or more screws to adjust a distance of the whole or a part of the hinge arm in respect to the mounting piece. Adjustment of the position and/or orientation of the hinge arm in respect to the mounting piece serves to adjust the furniture hinge or the furniture door or flap in respect to the furniture body or an adjacent furniture door or flap of the piece of furniture.

(8) The hinge assembly may comprise one or more hinge levers attached to the hinge cup on one side and to the hinge arm on the other side in an articulated manner. Several hinge levers preferably extend parallel or slightly oblique to each other. The hinge levers may have a flat extension or, in a longitudinal section, a straight extension, or, in a longitudinal section, they may have a curved or bent extension. The hinge assembly connects the components of the furniture hinge associated with the hinge cup to the components of the furniture hinge associated with the hinge arm.

(9) According to the invention, the cylinder of the damping element may be pivotally attached to a component of the furniture hinge which is arranged on a side of the hinge assembly facing the hinge arm, i.e. is associated with the hinge arm, or is arranged on a side of the hinge assembly

facing the hinge cup, i.e. is associated with the hinge cup. Another component of the furniture hinge, which is arranged on the respective other side of the hinge assembly, or a component of the hinge assembly itself, can act on the piston rod and provide for actuation of the damping element.
(10) A pivotable attachment of the cylinder of the damping element to a component of the furniture hinge other than the mounting piece has several advantages. On the one hand, the cylinder can change its position relative to the component to which it is pivotally attached during an opening movement and/or closing movement of the furniture hinge. The change in position of the cylinder can be defined, for example, by the movement of another component of the furniture hinge which is attached to the piston rod of the damping element for actuating the damping element. This ensures that a majority of the forces acting on the piston rod of the damping element during the opening movement and/or closing movement of the furniture hinge are always directed parallel to the direction of movement of the piston in the cylinder of the damping element. Transverse forces acting on the piston rod of the damping element during the opening movement and/or closing movement are significantly reduced, if not completely eliminated. As a result, the mechanical stress on the damping element, in particular on a seal of the piston, on the piston itself and/or on a bearing of the piston or the piston rod in the cylinder, is significantly reduced, which results in a longer service life of the damping element.
(11) Furthermore, in the case of furniture hinges in which the hinge arm is adjustably fastened to the mounting piece, the invention has the advantage that an adjustment of the position and/or orientation of the hinge arm in respect to the mounting piece does not result in a variation of an articulation point on the piston rod of the damping element in respect to the pivot axis via which the cylinder of the damping element is pivotably fastened to the one hinge part or the first component of the furniture hinge. In this way, an undesired variation of the damping properties or damping characteristics of the damping element can be avoided.
(12) According to an advantageous further development of the present invention, it is suggested that the cylinder of the damping element is fastened to the one hinge part or to the first component of the furniture hinge rigidly connected to the one hinge part pivotably about the pivot axis in such a manner that a pivot position of the cylinder of the damping element in respect to the one hinge part or to the first component of the furniture hinge rigidly connected to the one hinge part changes during the opening movement and/or closing movement of the furniture hinge. During the entire opening movement and/or closing movement between a fully closed position of the furniture hinge and a fully open position of the furniture hinge, the pivot position of the cylinder preferably changes continuously. However, it is also conceivable that the cylinder is pivotably fastened to the one hinge part or to the first component in such a way that during the entire opening movement and/or closing movement, starting from an initial position of the furniture hinge, the pivot positon of the cylinder increases and reaches a maximum value after part of the opening movement and/or closing movement, and that the pivot position then decreases again in the further course of the opening movement and/or closing movement until the furniture hinge reaches an end position.
(13) According to a preferred embodiment of the present invention, it is suggested that that hinge part to which the cylinder of the damping element is not fastened, or a second component of the furniture hinge connected to that hinge part, is connected to the piston rod of the damping element at least towards the end of the opening movement and/or the closing movement and actuates the damping element. The second component connected to that hinge part may be rigidly attached to that hinge part or movably connected thereto. Thus, it would be conceivable that the second component is formed by a component of the hinge assembly, for example a hinge lever. In this respect, it is advantageous that the furniture hinge does not require any additional components that form a deflection gear for straight-line actuation of the damping element. Instead, one of the components that is already present in a furniture hinge (e.g. the hinge cup or a hinge lever of the hinge assembly) acts directly on the piston rod and realises a direct actuation of the damping element.

(14) The component of the furniture hinge that actuates the damping element must move in respect to the component of the furniture hinge to which the cylinder of the damping element is pivotally attached during at least part of the opening movement and/or the closing movement of the furniture hinge. The component of the furniture hinge which actuates the damping element can actuate the piston rod of the damping element simply by abutment or contact. Preferably, however, it is suggested that the component of the furniture hinge which actuates the damping element is hinged to the piston rod, i.e. pivotably attached, so that the actuating component is in operative connection with the piston rod at all times during the opening movement and/or the closing movement of the furniture hinge. This has the further advantage that the damping element can be actuated both by pulling and by pushing on the piston rod. This enables damping of the furniture hinge both during the opening movement and during the closing movement.
(15) Advantageously, it is suggested that the hinge assembly comprises at least one hinge lever which is connected to the first hinge part pivotable about a first hinge axis and connected to the second hinge part pivotable about a second hinge axis, wherein the hinge lever forms the second component of the furniture hinge which is connected to the piston rod of the damping element at least towards the end of the opening movement and/or the closing movement and actuates the damping element. The hinge lever is mounted on the one hand on the first hinge axis of the hinge cup and on the other hand on the second hinge axis of the hinge arm. The hinge lever can have a flat or, in a longitudinal section, straight extension. Preferably, however, the hinge lever is curved or angled. The hinge lever may even be designed as an angled lever, wherein preferably the first hinge axis extends at one end of a first lever arm and the second hinge axis extends at or near a bending point or edge of the angled lever. The second lever arm, which is angled in respect to the first lever arm, can actuate the piston rod of the damping element.
(16) According to an alternative embodiment, it is suggested that the hinge assembly comprises at least two hinge levers, wherein a first hinge lever is connected to the first hinge part pivotable about a first hinge axis and connected to the second hinge part pivotable about a second hinge axis, and a second hinge lever is connected to the first hinge part pivotable about a third hinge axis and connected to the second hinge part pivotable about a fourth hinge axis, wherein the first hinge lever forms the second component of the furniture hinge which is connected to the piston rod of the damping element at least towards the end of the opening movement and/or the closing movement and actuates the damping element. Preferably, the first and third hinge axes as well as the second and fourth hinge axes are spaced apart from one another. The first hinge lever forms the second component of the furniture hinge which is connected to the piston rod of the damping element at least towards the end of the opening movement and/or the closing movement and actuates the damping element. The first hinge lever is mounted on the one hand on the first hinge axis of the hinge cup and on the other hand on the second hinge axis of the hinge arm. The second hinge lever is mounted on the one hand on the third hinge axis of the hinge cup and on the other hand on the fourth hinge axis of the hinge arm. The hinge levers can have a planar, i.e. in a longitudinal section a straight, extension. Preferably, however, at least one of the hinge levers is curved or bent in a longitudinal section. At least the first hinge lever is preferably designed as an angled lever. In this case, the first hinge axis preferably runs at one end of a first lever arm and the second hinge axis runs at a bending point or edge of the angled lever. The second lever arm of the angle lever, which is angled in respect to the first lever arm, can actuate the piston rod of the damping element.
(17) Preferably, the first hinge axis and the third hinge axis are arranged on the hinge cup at a first distance from each other. Likewise, preferably, the second hinge axis and the fourth hinge axis are arranged on the hinge arm at a second distance from each other. The first and second distances may be equal, but preferably are different.
(18) The hinge axes may be formed by holes or eyelets formed in the at least one hinge arm, through which hinge shafts are guided. The shafts are mounted in the hinge cup, the hinge arm or in another component of the furniture hinge connected to the hinge cup or the hinge arm, and thus

ensure articulation of the at least one hinge lever in respect to the corresponding components (i.e. hinge cup, hinge arm or component connected thereto) of the furniture hinge.

(19) According to another advantageous embodiment of the present invention, it is suggested that hinge part to which the cylinder of the damping element is not fastened, or a second component of the furniture hinge connected to that hinge part, is fastened to the piston rod of the damping element pivotable about an articulation axis. If the damping element is actuated, for example, by means of a hinge lever of the hinge assembly, the hinge lever could be pivotably articulated to the piston rod via the articulation axis. The articulation axis can, for example, be formed by a hole formed in the piston rod, through which an articulation shaft is guided, which is mounted in the hinge lever. For this purpose, the hinge lever can have one or more eyelets or one or more holes through which the articulation shaft is guided. An eyelet can be formed, for example, by bending one end of a lever arm of the hinge lever around a bending axis running transversely to the longitudinal extension of the hinge lever. A hole may be formed, for example, by drilling or punching in one or more lateral tabs of a lever arm of the hinge lever, wherein the lateral tabs may be formed by bending lateral portions of the lever arm about a bending axis extending parallel to the longitudinal extension of the hinge lever.

(20) According to a further preferred embodiment of the present invention, it is suggested that the second component of the furniture hinge is fastened to the piston rod of the damping element pivotable about an articulation axis, wherein a surface of the furniture body to which the mounting piece is attached defines a first plane, wherein a second plane is defined which extends parallel to the first plane and passes through the second hinge axis, wherein a third plane is defined which extends parallel to the first plane and passes through the fourth hinge axis, and wherein the articulation axis is located between the second plane and the third plane throughout the entire opening movement and/or closing movement of the furniture hinge. This design of the furniture hinge allows for a particularly compact construction.

(21) Advantageously, in a fully closed position of the furniture hinge, the articulation axis is arranged close to a straight line connecting the second hinge axis with the fourth hinge axis, and a distance of the articulation axis to the connecting straight line increases continuously during the entire opening movement of the furniture hinge. Accordingly, the pivot angle of the cylinder of the damping element about the pivot axis relative to the component of the furniture hinge to which the cylinder is pivotably fastened also changes continuously during the entire opening movement of the furniture hinge.

(22) According to a further advantageous embodiment of the invention, it is suggested that the second hinge part is movably fastened to the mounting piece attached to the furniture body and can be fixed to the mounting piece in a relative position in respect to the mounting piece, so that by changing the relative position between the second hinge part and the mounting piece the furniture door or flap can be adjusted in at least one direction relative to the furniture body or to an adjacent door or flap of the piece of furniture. The second hinge part is preferably the hinge arm.

(23) For example, the hinge arm may be held to the mounting piece by means of at least one screw. By loosening the screw, a longitudinal displaceability of the hinge arm with respect to the mounting piece in a direction parallel to a longitudinal extension of the hinge arm could be released. By longitudinally displacing the hinge arm (together with the components of the furniture hinge and the furniture door or flap attached thereto), an offset of the closed furniture door or flap in respect to the furniture body or an adjacent door or flap of the piece of furniture can be adjusted. When the offset is set correctly, the at least one screw can be tightened again so that the hinge arm is fixed to the mounting piece in the respective position relative to the mounting piece.

(24) Furthermore, it is conceivable to adjust a distance of at least a part of the hinge arm in respect to the mounting piece by means of another screw which is guided in the hinge arm and supported on the mounting piece. By turning this screw, the closed furniture door or flap can be displaced in respect to the furniture body or an adjacent door or flap of the piece of furniture along its surface

extension. This can be used, for example, to adjust a gap width between the furniture door or flap and the furniture body or an adjacent door or flap of the piece of furniture. If the door or flap is attached to the furniture body by means of a plurality of furniture hinges, the door or flap can also be raised if only the distance of the lower furniture hinge is increased and/or that of the upper furniture hinge is reduced. Similarly, by turning the screw in the opposite direction and/or by increasing the distance of the upper furniture hinge, the door or flap can also be lowered.

(25) Another advantageous embodiment of the invention provides that the second hinge part is releasably attached to the mounting piece attached to the furniture body, so that by releasing the attachment of the second hinge part to the mounting piece, the furniture door or flap is releasable from the furniture body. The releasable or detachable attachment of the second hinge part to the mounting piece can be realised in different ways. Various embodiments of a releasable attachment of a hinge arm to a mounting piece are known from the prior art. Preferably, the hinge arm has a spring-loaded movable latching element which, when the hinge arm is placed on the mounting piece, engages behind a section of the mounting piece and secures the hinge arm to the mounting piece. The latching element is preferably linearly movable, in particular in the direction of the longitudinal extension of the hinge arm. A spring force presses the latching element into the engaging position and prevents unintentional release of the hinge arm from the mounting piece. To release the hinge arm from the mounting piece, the latching element must be actuated against the spring force, e.g. by a user's finger or a suitable tool, so that it no longer engages behind the section of the mounting piece. Preferably, when the hinge arm is placed on the mounting piece for attachment, the latching element automatically and self-actuates into the engaging position.

(26) A releasable attachment of the hinge arm to the mounting piece is possible in the present invention, since the cylinder of the damper element is not attached to the mounting piece, but instead to a component of the furniture hinge, e.g. the hinge arm or the hinge cup or a component fixedly attached thereto, which is detached from the mounting piece together with the hinge arm. In the invention, by detaching the hinge arm from the mounting piece, the complete furniture hinge, except for the mounting piece, is separated from the furniture body. Previously made settings and proper functioning of the furniture hinge are fully retained, so that after the hinge arm is placed and fastened to the mounting piece again, the furniture hinge is immediately fully functional again. This applies in particular to the function of the damping element.

(27) Advantageously, the furniture hinge comprises at least one spring element which supports the opening movement and/or the closing movement at least towards the end of the movement and urges the furniture door or flap into an open position and/or a closed position in respect to the furniture body. The furniture hinge according to the invention thus combines the function of a damping hinge with that of a spring hinge. Preferably, the spring element is formed as a torsion spring made of metal, in particular of steel. This allows a particularly compact design of the furniture hinge despite its multifunctionality. The spring element is preferably mounted pre-tensioned in the hinge and presses the furniture door or flap into the closed position. Preferably, the spring element on the one hand leans on the second hinge part (i.e. the hinge arm) or on a third component of the furniture hinge connected to the second hinge part and on the other hand on the first hinge part (i.e. the hinge cup) or on a fourth component of the furniture hinge connected to the first hinge part. For example, it is conceivable that the spring element is supported on the one hand by the hinge arm and on the other hand by a hinge lever of the hinge assembly, preferably by the first hinge lever in the case of a multiple-lever hinge assembly. In particular, it is suggested that the spring element is supported, on the other hand, by the first hinge axis via which the first hinge lever is articulated to the hinge cup, or on a corresponding hinge shaft which is guided through holes in side walls of the hinge cup and holes or an eyelet in the first hinge lever. The hinge shaft is thus rigidly connected to the hinge cup. It is also possible that on the other hand, the spring element is supported on the first hinge axis or the corresponding hinge shaft during a first part of an opening movement and/or closing movement of the furniture hinge, and on the third hinge axis or the

corresponding hinge shaft during a further part of the opening movement and/or closing movement. The second hinge lever is articulated to the hinge cup by means of the third hinge axis. The hinge shaft corresponding to the third hinge axis is guided through holes in the side walls of the hinge cup and holes or an eyelet in the second hinge lever. The hinge shaft is thus rigidly connected to the hinge cup.

(28) The pivotable attachment of the cylinder of the damping element to one of the hinge parts or to the first component of the furniture hinge, which is rigidly connected to one of the hinge parts, can be realised in different ways. Preferably, a bearing element is arranged on an outer circumferential surface of the cylinder of the damping element between the ends of the cylinder, by means of which bearing element the cylinder of the damping element is pivotably fastened to the second hinge part or to the first component of the furniture hinge rigidly connected to the second hinge part. The bearing element may be integrally formed in one piece with the cylinder. Alternatively, the bearing element could be a separate part that is attached to the outer circumferential surface of the cylinder, for example by means of welding, soldering, gluing, riveting, screwing, etc. Preferably, the bearing element has two bearing parts of the same type, with one bearing part arranged on each side of a vertical centre plane through the cylinder along its longitudinal axis. This allows symmetrical loading of the damping element when force is applied to the piston rod during the opening movement and/or closing movement of the furniture hinge. Each bearing part may be provided with a hole through which a pivot shaft is guided. The pivot shaft may further extend laterally through respective holes provided in side walls of the hinge arm and secured in respect to the hinge arm.

(29) Preferably, the pivot axis of the pivotable attachment of the cylinder of the damping element to the hinge part or the first component attached thereto, extends at a distance from a longitudinal axis of the cylinder of the damping element. If the cylinder is pivotably attached to the second hinge part, the pivot axis of the pivotable attachment of the cylinder to the second hinge part extends at a distance from a longitudinal axis of the cylinder of the damping element. The pivot axis runs essentially centrally between the two ends of the cylinder. The pivotable attachment is preferably located essentially in the middle between the two opposite ends of the cylinder. This causes a relatively small pivoting movement of the cylinder of the damping element during the opening movement and closing movement of the furniture hinge, whereby the size and the required installation space for the furniture hinge can be further reduced.

(30) Further features and advantages of the present invention are explained in more detail below with reference to the figures, which show preferred embodiments of the invention. It is noted that the features shown in the figures and described below may each be essential to the invention also on their own, even if not expressly shown in the figures and not expressly described in the description. Furthermore, the features shown in the individual figures can be combined with each other in any possible way, even if the features belong to different embodiments and the combination is not explicitly mentioned in the description. Finally, one or more of the features shown in the figures and described in the description may also be omitted if they are not necessary for the realisation of the invention. The figures show:

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** a first embodiment of the furniture hinge according to the invention in an exploded view;

(2) FIG. **2** the furniture hinge of FIG. **1** in an assembled state in a fully open position in a perspective view;

(3) FIG. **3** the furniture hinge of FIG. **1** in an assembled state in a fully closed position in a partial

longitudinal sectional side view;
(4) FIG. **4** the furniture hinge of FIG. **2** in a partial longitudinal sectional perspective view;
(5) FIG. **5** the furniture hinge of FIG. **4** in an intermediate position;
(6) FIG. **6** the furniture hinge of FIG. **4** in a fully closed position;
(7) FIG. **7** a second embodiment of the furniture hinge according to the invention in an exploded view;
(8) FIG. **8** the furniture hinge of FIG. **7** in an assembled state in a fully open position in a perspective view;
(9) FIG. **9** the furniture hinge of FIG. **8** in a partial longitudinal sectional perspective view;
(10) FIG. **10** the furniture hinge of FIG. **9** in an intermediate position; and
(11) FIG. **11** the furniture hinge of FIG. **9** in a fully closed position.

DETAILED DESCRIPTION

(12) A furniture hinge according to a first preferred embodiment is shown in FIGS. **1** to **6** and designated in its entirety with reference sign **2**. The furniture hinge **2** may serve for fastening a furniture door **4** or flap of a piece of furniture **6** to a furniture body **8** of the piece of furniture **6** in an articulated manner (see FIG. **3**). The furniture hinge **2** comprises a first hinge part **10** for attachment to the furniture door **4** or flap, and a second hinge part **12** for attachment to the furniture body **8**.

(13) The first hinge part **10** may be designed as a hinge cup, which can be placed in a recess on the inside of the furniture door **4** or flap and fastened thereto in a known manner, for example by means of screws (not shown). The screws are guided through holes **14** which are provided in a collar **16** attached to or making an integral part of the hinge cup **10**. The collar **16** lies on the inside surface of the door **4** or flap surrounding the recess. The second hinge part **12** may be designed as a hinge arm, which may be directly or indirectly attached to the furniture body **8** on the inside of the piece of furniture **6**. The hinge arm **12** can be attached to the furniture body **8** in an adjustable manner and/or in an easily releasable manner, e.g. by means of a snap-in connection, which will be described in more detail below.

(14) The furniture hinge **2** further comprises a hinge assembly **18** (**18.1** and **18.2**) which movably connects the first hinge part **10** to the second hinge part **12** in order to realise an opening movement of the furniture hinge **2**, in which the first hinge part **10** moves away from the second hinge part **12**, and an oppositely directed closing movement of the furniture hinge **2**. The hinge assembly **18** may comprise one or more hinge levers **18.1**, **18.2** on one side attached to the hinge cup **10** and on the opposite side to the hinge arm **12** in an articulated manner. Several hinge levers **18.1**, **18.2** preferably extend parallel or slightly oblique to each other. The hinge levers **18.1**, **18.2** may have a flat extension or, in a longitudinal section, a straight extension (see second hinge lever **18.2**). Alternatively, in a longitudinal section, they may have a curved or bent extension (see first hinge lever **18.1**). The hinge assembly **18** connects the components of the furniture hinge **2** associated with the hinge cup **10** to the components of the furniture hinge **2** associated with the hinge arm **12**.

(15) In the shown embodiments, the hinge assembly **18** comprises two hinge levers **18.1**, **18.2**. A first hinge lever (internal hinge lever) **18.1** is connected to the first hinge part **10** pivotable about a first hinge axis **20.1** and connected to the second hinge part **12** pivotable about a second hinge axis **20.2**. A second hinge lever **18.2** is connected to the first hinge part **10** pivotable about a third hinge axis **20.3** and to the second hinge part **12** pivotable about a fourth hinge axis **20.4**. The first hinge axis **20.1** and the third hinge axis **20.3** are spaced apart from each other by a first distance. The second hinge axis **20.2** and the fourth hinge axis **20.4** are spaced apart from each other by a second distance. Preferably, the two distances are different from each other.

(16) In the shown embodiments, the hinge axes **20** are realized by means of hinge shafts **78** which extend through holes provided in side walls **80** of the first hinge part **10** or in side walls **82** of the second hinge part **12**. In particular, a first hinge shaft **78.1** extends through holes in the side walls **80** of the hinge cup **10** and through respective holes or an eyelet **83** in the first hinge lever **18.1**

defining the first hinge axis **20.1**, a second hinge shaft **78.2** extends through holes in the side walls **82** of the hinge arm **12** and through respective holes or an eyelet in the first hinge lever **18.1** defining the second hinge axis **20.2**, a third hinge shaft **78.3** extends through holes in the side walls **80** of the hinge cup **10** and respective holes or an eyelet in the second hinge lever **18.2** defining the third hinge axis **20.3**, and a fourth hinge shaft **78.4** extends through holes in the side walls **82** of the hinge arm **12** and respective holes or an eyelet in the second hinge lever **18.2** defining the fourth hinge axis **20.4**. In the embodiments the first and third hinge shafts **78.1**, **78.3** are attached to each other forming a single U-shaped hinge element **78.5**.
(17) Furthermore, the furniture hinge **2** comprises at least one damping element **22** for damping the opening movement and/or the closing movement of the furniture hinge **2** at least towards the end of the movement. The damping element **22** comprises an external cylinder **24**, a piston (not visible) which is linearly movable inside the cylinder **24** relative thereto, and a piston rod **26** which is coupled to the piston and projects from the cylinder **24**.
(18) Finally, the furniture hinge **2** of the first embodiment comprises a plate-like mounting piece **27** which is attached to the furniture body **8** and to which the second hinge part **12** may be attached. For example, the mounting piece **27** may be attached to the furniture body **8** by means of screws, which pass through holes **28** provided in the mounting piece **27**. Other ways of attachment of the mounting piece **27** to the furniture body **8** are conceivable, too. The hinge arm **12** may be attached to the mounting piece **27** directly or indirectly in a rigidly fixed manner or in an adjustable manner and/or in a releasable manner.
(19) In the first embodiment of FIGS. **1** to **6**, the second hinge part **12** is attached to the mounting piece **27** not directly but rather by means of an intermediate piece **30**. The intermediate piece **30** provides for a releasable attachment of the hinge arm **12** to the mounting piece **27**, which will be described in further detail below. Furthermore, the hinge arm **12** is adjustable in respect to the intermediate piece **30**, in order to adjust the position, offset and/or orientation of the closed door **4** or flap in respect to the furniture body **8** or an adjacent door or flap of the piece of furniture **6**.
(20) In order to realize the adjustment of the furniture hinge **2**, the hinge arm **12** can be attached to the intermediate piece **30**, e.g. by means of a fastening screw **32**. The screw **32** is guided through a guiding slot **34** provided in a top wall of the hinge arm **12** and then screwed into a threaded hole **36** provided in a top surface of the intermediate piece **30**. The slot **34** allows a movement of the hinge arm **12** in respect to the intermediate piece **30** in a longitudinal direction **38** when the screw **32** is slightly loosened. By tightening the screw **32** the relative position of the hinge arm **12** in respect to the intermediate piece **30** is fixed.
(21) Furthermore, an adjustment screw **40** is provided which is screwed into a threaded hole **42** provided in the top surface of the hinge arm **12** so that it protrudes into the inside of the hinge arm **12** (see FIG. **3**). A bottom distal end of the adjustment screw **40** has a collar **44** which is separated from a threaded section **46** of the screw **40** by a necked region **48** (see FIG. **3**). When mounting the hinge arm **12** onto the intermediate piece **30**, the necked region **48** is inserted into a slot **50** provided in the top surface of the intermediate piece **30**. After the hinge arm **12** is mounted to the intermediate piece **30** and the screw **32** is fastened, the adjustment screw **40** may be actuated, i.e. turned in one direction or the other, thereby increasing or decreasing a distance of the top surface of the hinge arm **12** in respect to the intermediate piece **30**.
(22) Thus, by turning the adjustment screw **40**, the closed furniture door **4** or flap can be displaced in respect to the furniture body **8** or an adjacent door or flap of the piece of furniture **6**, the displacement being effected parallel to the surface extension of the door **4** or flap. This can be used, for example, to adjust a gap width between the furniture door **4** or flap and the furniture body **8** or an adjacent door or flap of the piece of furniture **6**. If the door **4** or flap is attached to the furniture body **8** by means of a plurality of furniture hinges **2**, the door **4** or flap can also be raised if only the distance between hinge arm **12** and intermediate piece **30** of the lower furniture hinge **2** is increased and/or that of the upper furniture hinge **2** is reduced. Similarly, by turning the adjustment screw **40**

in the opposite direction and or by increasing the distance of the upper furniture hinge **2** or reducing the distance of the lower furniture hinge **2**, the door **4** or flap can also be lowered.
(23) With simple words, in this embodiment the hinge arm **12** comprises two separate elements (i.e. hinge arm **12** and intermediate piece **30**), which are attached to each other adjustable in respect to each other in the longitudinal direction **38** (when the screw **32** is loosened) and which can be fixed to each other in the adjusted relative position (when the screw **32** is tightened). Further, the adjustment screw **40** serves for adjustment of the distance between the two separate components **12** and **30** in a direction perpendicular to the longitudinal direction **38**.
(24) In order to realize the releasable attachment of the hinge arm **12** to the mounting piece **27** and to the furniture body **8**, respectively, the intermediate piece **30** (with the hinge arm **12** attached thereto) is releasably attached to the mounting piece **27**. This is achieved by a simple snap-in connection. To this end, it is suggested that the intermediate piece **30** comprises front hook-like elements **52**, which are adapted to engage behind a first front holding section **54** of the mounting piece **27**. Further, the intermediate piece **30** has positioning elements **56** facing towards the mounting piece **27**, which are adapted to engage with a respective receiving section **58** of the mounting piece **27**. The receiving section **58** preferably comprises slots or holes in the top surface of the mounting piece **27** into which the positioning elements **56** can be inserted.
(25) A spring-loaded release element **60** is slidably guided in the intermediate piece **30**. In this embodiment, the release element **60** is a push-button which is linearly movable in respect to the intermediate piece **30** in the longitudinal direction **38**. Of course, the release element **60** could be designed differently from what is shown in FIGS. **1** to **6**. The push-button **60** is pressed to the rear of the intermediate piece **30** by means of a pressure spring **62**. As can be seen from FIG. **3**, the push-button **60** has at least one, preferably two hook-like snap elements **64** which are adapted to engage with a second rear holding section **66** of the mounting piece **27**. The snap elements **64** each have a slanted sliding surface **65** facing downwards towards the mounting piece **27** during attachment of the intermediate piece **30** to the mounting piece **27**. When the intermediate piece **30** is lowered onto the mounting piece **27**, the sliding surfaces **65** slide over the second holding section **66** and urge the push-button **60** forward overcoming the force of spring element **62** and letting the snap elements **64** engage with the second holding section **66** of the mounting piece **27**. After engagement of the snap elements **64**, the spring **62** presses and holds the push-button **60** in the engaged rear position. Further, the push-button **60** may have a recess **68** which opens into the top surface of the push-button **60** and which is adapted to receive a distal end with a threaded section of the fastening screw **32**. Finally, the push-button **60** may be provided with an actuating surface **70** for actuating the push-button **60** by a user, e.g. with one of his fingers or a tool.
(26) For attachment of the hinge arm **12** (and of all the other components of the furniture hinge **2** connected to the hinge arm **12**, including the intermediate piece **30**, the hinge assembly **18**, the hinge cup **10** and the door **4** or flap) to the mounting piece **27**, the hook-like elements **52** are brought into engagement with the first holding section **54** of the mounting piece **27**. Then the rear part of the hinge arm **12** is swivelled downwards about an imaginary axis defined by the coupling of the hook-like elements **52** and the first holding section **54** and the positioning elements **56** are inserted into the receiving holes **58**. Finally, the at least one snap element **64** of the push-button **60** is automatically brought into engagement with the second holding section **66** of the mounting piece. **27**.
(27) For detachment of the hinge arm **12** (and of all the other components of the furniture hinge **2** connected to the hinge arm **12**) from the mounting piece **27**, the push-button **60** is pressed forward against the force of the pressure spring **62**. This results in a releasement of the engagement between the hook-like elements **64** and the second holding section **66** of the mounting piece **27**. The rear part of the hinge arm **12** can be swivelled upwards about the imaginary axis defined by the coupling of the hook-like elements **52** and the first holding section **54** and then the hinge arm **12** can be moved forward thereby releasing the engagement between the hook-like elements **52** and

the first holding section **54** of the mounting piece **27**. Finally, the hinge arm **12** and all the other components of the furniture hinge **2** connected to the hinge arm **12** can be separated from the mounting piece **27** and from the furniture body **8**.
(28) Preferably, the furniture hinge **2** comprises at least one spring element **72** which supports the opening movement and/or the closing movement at least towards the end of the movement and urges the furniture door **4** or flap into an open position and/or a closed position in respect to the furniture body **8**. In the shown embodiment, the furniture hinge **2** comprises one spring element **72** designed as a torsion spring, which supports the closing movement. A torsion spring allows a particularly compact design of the furniture hinge **2** despite its multifunctionality (functioning as a damping hinge and a spring hinge). The torsion spring **72** can be made of metal, in particular of steel. The torsion spring **72** can be wound around the fourth hinge shaft **78.4** and thereby positioned inside the hinge arm **12**. The spring element **72** is preferably mounted in a pre-tensioned manner in the hinge **2** and urges and holds the furniture door **4** or flap in its closed position.
(29) Preferably, on the one hand the spring element **72** leans with its first end **74** on the hinge arm **12** or on a third component of the furniture hinge **2** connected to the hinge arm **12**, and on the other hand with its second end **76** on the hinge cup **10** or on a fourth component of the furniture hinge **2** connected to the hinge cup **10**. For example, it is conceivable that the spring element **72** is supported with its first end **74** by the hinge arm **12** and/or by a support shaft **84** which is guided through holes in the side walls **82** of the hinge arm **12** and thereby fixedly connected to the hinge arm **12**. The support shaft **84** provides for a steady connection of the spring element **72** to the hinge arm **12**. It is further conceivable that the spring element **72** is supported with its second end **76** by one or more of the hinge levers **18.1**, **18.2** of the hinge assembly **18**, which are connected to the hinge cup **10** in an articulated manner, or by one or more of the hinge shafts **78.1**, **78.3**, which are fixedly connected to the hinge cup **10** and form the first and third hinge axes **20.1**, **20.3** of the first and second hinge levers **18.1**, **18.2**, respectively. In particular, it is suggested that the second end **76** of the spring element **72** is supported by the U-shaped hinge element **78.5**. Thus, in that case the second end **76** of the spring element **72** is supported by and slides along the first hinge shaft **78.1** during a first part of an opening movement and/or closing movement of the furniture hinge **2**, and is supported by and slides along the third hinge shaft **78.3** during a further part of the opening movement and/or closing movement. This has the advantage that depending on the current opening or closing angle of the door **4** or flap in respect to the furniture body **8**, different spring forces act on the hinge cup **10** and the door **4** or flap, respectively.
(30) According to the present invention, the cylinder **24** of the damping element **22** is fastened to one of the hinge parts **10** or **12** or to a first component of the furniture hinge **2**, which is connected to one of the hinge parts **10** or **12** at least during the opening movement and/or closing movement of the furniture hinge **2**, pivotably about a pivot axis **86**. In the shown embodiment, the cylinder **24** is fastened to the hinge arm **12** in a pivotable manner. This is realized by means of a pivot shaft **88**, which is guided through holes in the side walls **82** of the hinge arm **12**. Of course, alternatively, the damping element **22** could also be fastened to the hinge cup **10** or to the intermediate piece **30** pivotable about a pivot axis.
(31) The pivotable attachment of the cylinder **24** of the damping element **22** to a component of the furniture hinge **2** other than the mounting piece **27** has several advantages. On the one hand, the cylinder **24** can change its pivot position relative to the component to which it is pivotally attached during an opening movement and/or closing movement of the furniture hinge **2**. The change in position of the cylinder **24** can be defined, for example, by the movement of another component of the furniture hinge **2** which is attached to the piston rod **26** of the damping element **22** for actuating the damping element **22**. This ensures that a majority of the forces acting on the piston rod **26** during the opening movement and/or closing movement of the furniture hinge **2** are always directed parallel to the direction of movement of the piston in the cylinder **24** of the damping element **22**. Transverse forces acting on the piston rod **26** of the damping element **22** during the opening

movement and/or closing movement are significantly reduced, if not completely eliminated. As a result, the mechanical stress on the damping element **22**, in particular on a seal of the piston, on the piston itself and/or on a bearing of the piston or the piston rod **26** in the cylinder **24**, is significantly reduced, which results in a longer service life of the damping element **22**.

(32) Furthermore, in the case of a furniture hinge **2** in which the hinge arm **12** is adjustably fastened to the mounting piece **27**, the invention has the advantage that an adjustment of the position and/or orientation of the hinge arm **12** in respect to the mounting piece **27** does not result in a variation of the pivot axis **86** via which the cylinder **24** of the damping element **22** is pivotably fastened to the one hinge part **10** or **12** in respect to an articulation point (running through an articulation axis **90**) on the piston rod **26** of the damping element **22**. In this way, an undesired variation of the damping properties or damping characteristics of the damping element **22** due to adjustment of the hinge arm **12** in respect to the mounting piece **27** can be avoided.

(33) Another component of the furniture hinge **2**, which is arranged on the respective other side of the hinge assembly **18** in respect to the pivot axis **86**, or a component of the hinge assembly **18** itself, can act on the piston rod **26** in the articulation point and provide for actuation of the damping element **22**. Preferably, the other component acting on the piston rod **26** and actuating the damping element **22**, is attached to the piston rod **26** pivotably about the articulation axis **90**. To this end it is suggested that an articulation shaft **92** is inserted into the hinge arm **12** through holes **94** provided in the side walls **82** of the hinge arm **12** and guided through respective holes in the first hinge lever **18.1** and in the piston rod **26**. During operation of the furniture hinge **2**, the articulation shaft **92** is movable in respect to the hinge arm **12**. The holes in the first hinge lever **18.1** and in the piston rod **26** are aligned with the holes **94** in the side walls **82** of the hinge arm **12** when the furniture hinge **2** is in its completely opened position (see FIG. **2**). When the furniture hinge **2** is moved towards its closed position (see FIG. **3**), the holes in the first hinge lever **18.1** and in the piston rod **26** are increasingly misaligned in respect to the holes **94** in the side walls **82** of the hinge arm **2**.

(34) Preferably, a pivot position of the cylinder **24** of the damping element **22** in respect to the one hinge part **10** or **12**, to which the cylinder **24** is pivotably fastened, changes during the opening movement and/or closing movement of the furniture hinge **2**. The pivot position of the cylinder **24** preferably changes continuously during the entire opening movement and/or closing movement between a fully closed position of the furniture hinge **2** and a fully opened position of the furniture hinge **2**. However, it is also conceivable that the cylinder **24** is pivotably fastened to the one hinge part **10** or **12** in such a way that during the opening movement and/or closing movement, starting from an initial position of the furniture hinge **2**, the pivot positon of the cylinder **24** increases (or decreases) and reaches a maximum value after part of the opening movement and/or closing movement, and then decreases (or increases) again in the further course of the opening movement and/or closing movement until the furniture hinge **2** reaches an end position.

(35) Preferably, the other component of the furniture hinge **2** acting on the piston rod **26** and actuating the damping element **22**, is movably connected to the hinge cup **10**. In particular, it is suggested that the other component is formed by the first hinge lever **18.1**. In this respect, it is advantageous that the furniture hinge **2** does not require any additional components that form a deflection gear or the like for a straight-line actuation of the piston rod **26** of the damping element **22**. Instead, the first hinge lever **18.1** acts directly on the piston rod **26** and realises a direct actuation of the damping element **22**.

(36) In the described embodiments, the other component of the furniture hinge **2** (e.g. the first hinge lever **18.1**) which acts on the piston rod **26** of the damping element **22** during the entire opening movement and/or closing movement and actuates the damping element **22**, is pivotably attached to the piston rod **26**. This has the advantage that the damping element **22** can be actuated both by pulling and by pushing on the piston rod **26**. This enables damping of the furniture hinge **2** both during the opening movement and during the closing movement. While the second hinge lever **18.2** has an essentially planar extension, the first hinge lever **18.1** is preferably designed as an

angled lever. In this case, the first hinge axis **20.1** through the eyelet **83** runs at one end of a first lever arm and the second hinge axis **20.2** runs along a bending edge of the angled lever **18.1**. The second lever arm of the angled lever **18.1** extends in an angle in respect to the first lever arm and actuates the piston rod **26** of the damping element **22**. To this end, holes are provided at the distal end of the second lever arm through which the articulation axis **90** runs.
(37) The hinge axes **20** may be formed by holes or eyelets **83** formed in the hinge levers **18.1**, **18.2** and the respective holes in the side walls **80** of the hinge cup **10** or in the side walls **82** of the hinge arm **12**, respectively. The hinge shafts **78** are guided there through and mounted (fixed or attached) to the hinge cup **10** or hinge arm **12**, respectively. The shafts **78** are mounted in the respective holes in the side walls **80**, **82** of the hinge cup **10** or the hinge arm **12**, for example, by means of soldering, welding or by compression of the distal ends of the shafts **78** so that their diameter expands and they are clamped in the holes of the hinge cup **10** or the hinge arm **12**. Similarly, the pivot axis **86** is defined by holes in the cylinder **24** (or in a bearing element **104** attached thereto) and respective holes in the side walls **82** of the hinge arm **12**. The pivot shaft **88** is mounted in the respective holes in the side walls **82** of the hinge arm **12**, for example, by means of soldering, welding or by compression of the distal ends of the shaft **88**. Similarly, the articulation axis **90** is defined by holes in the distal end of the second lever arm of the first hinge lever **18.1** and a hole in the piston rod **26**. The articulation shaft **92** is mounted in the respective holes in the in the distal end of the second lever arm of the first hinge lever **18.1**, for example, by means of soldering, welding or by compression of the distal ends of the shaft **92**.
(38) A surface of the furniture body **8** to which the mounting piece **27** is attached defines an imaginary first plane **96** (see FIG. **3**), a second imaginary plane **98** is defined which extends parallel to the first plane **96** and passes through the second hinge axis **20.2**, and a third imaginary plane **100** is defined which also extends parallel to the first plane **96** and passes through the fourth hinge axis **20.4**. The articulation axis **90** is located between the second plane **98** and the third plane **100** throughout the entire opening movement and/or closing movement of the furniture hinge **2**. This design of the furniture hinge **2** allows for a particularly compact construction.
(39) Advantageously, in a fully closed position of the furniture hinge **2** (see FIG. **3**), the articulation axis **90** is arranged close to or even on a straight line **102** connecting the second hinge axis **20.2** with the fourth hinge axis **20.4**. A distance of the articulation axis **90** to the connecting straight line **102** increases continuously during the entire opening movement of the furniture hinge **2**. Accordingly, the pivot angle of the cylinder **24** of the damping element **22** about the pivot axis **86** in respect to the component of the furniture hinge **2** to which the cylinder **24** is pivotably fastened (e.g. the hinge arm **12**) changes continuously during the entire opening movement and the entire closing movement of the furniture hinge **2**.
(40) A releasable attachment of the hinge arm **12** to the mounting piece **27** is possible in the present invention, since the cylinder **24** of the damper element **22** is not attached to the mounting piece **27**, but instead to a hinge part **10**, **12** of the furniture hinge **2** or to the first component fixedly attached thereto, which will be detached from the mounting piece **27** together with the hinge arm **12**. In the invention, by detaching the hinge arm **12** from the mounting piece **27**, the complete furniture hinge **2**, except for the mounting piece **27**, is separated from furniture body **8**. Previously made settings and proper functioning of the furniture hinge **2** are fully retained, so that after the hinge arm **12** is placed back and fastened to the mounting piece **27** again, the furniture hinge **2** is immediately fully operable again. This applies in particular to the function of the damping element **22**.
(41) The pivotable attachment of the cylinder **24** of the damping element **22** to one of the hinge parts **10**, **12** or to the first component of the furniture hinge **2**, which is rigidly connected to one of the hinge parts **10**, **12**, can be realised in different ways. Preferably, the bearing element **104** is arranged on an outer circumferential surface of the cylinder **24** of the damping element **22** between the ends of the cylinder **24**. By means of the bearing element **104** the cylinder **24** is pivotably fastened to the second hinge part **12** or to the first component of the furniture hinge **2**. The bearing

element **104** may be integrally formed in one piece with the cylinder **24**. Alternatively, the bearing element **104** could be a separate part that is attached to the outer circumferential surface of the cylinder **24**, for example by means of welding, soldering, gluing, riveting, screwing, etc.
(42) Preferably, the bearing element **104** has two bearing parts of the same type, with one bearing part arranged on each side of a vertical centre plane through the cylinder **24** along its longitudinal axis. The bearing parts each have an essentially planar extension parallel to each other. The two bearing parts allow symmetrical loading of the damping element **22** when force is applied to the piston rod **26** during the opening movement and/or closing movement of the furniture hinge **2**. Each bearing part of the bearing element **104** may be provided with a hole through which the pivot shaft **88** is guided.
(43) Preferably, the hinge cup **10**, the hinge arm **12**, the intermediate piece **30**, the mounting piece **27** and the hinge levers **18**.**1**, **18**.**2** are made of metal, in particular steel. It is further suggested that the hinge cup **10** and the mounting piece **27** are manufactured from a sheet metal by means of a punching process and a subsequent deep-drawing process. The hinge arm **12**, the intermediate piece **30**, and the hinge levers **18**.**1**, **18**.**2** are preferably manufactured from a sheet metal by means of a punching process and a subsequent bending process. By bending lateral portions of the punched-out sheet metal along a bending axis extending parallel to the longitudinal direction **38** the side walls **82** of the hinge arm **12**, side walls of the intermediate piece **30** and lateral tabs of the hinge levers **18**.**1**, **18**.**2** can be formed. Thus, for example, the hinge arm **12** comprises two side walls **82** extending parallel to each other and each comprising the holes for defining the second and fourth hinge axes **20**.**2**, **20**.**4**, the pivot axis **86**, and for inserting the support shaft **84** and the articulation shaft **92**. Similarly, the second lever arm of the first hinge lever **18**.**1** comprises two lateral tabs extending parallel to each other and each comprising the holes for defining the second hinge axis **20**.**2** and the articulation axis **90**. Similarly, the second hinge lever **18**.**2** comprises two lateral tabs extending parallel to each other and each comprising the holes for defining the third and fourth hinge axes **20**.**3**, **20**.**4**. Finally, the intermediate piece **30** comprises two side walls extending parallel to each other and defining two separate hook-like elements **52** and two separate positioning elements **56**. This provides for a symmetrical loading of the furniture hinge **2**.
(44) A top surface of the mounting piece **27** comprises a ditch-like depression **106** between the two front holding sections **54**, the depression **106** extending parallel to the longitudinal direction **38**. The depression **106** serves for receiving the cylinder **24** of the damping element **22** during its pivoting movement about pivot axis **86** during an opening movement and/or a closing movement of the furniture hinge **2**. This allows the design of a particular compact furniture hinge **2**.
(45) FIGS. **7** to **11** show a second preferred embodiment of the present invention. The main difference in respect to the first embodiment of FIGS. **1** to **6** is that the hinge arm **12** is directly attached to a dome-shaped mounting piece **27**′ and that there is no intermediate piece **30**. Thus, the hinge arm **12** cannot be released from the mounting piece **27**′ simply by releasing a snap-in connection. Rather, the screws **32** and **40** would have to be loosened in order to separate the hinge arm **12** from the mounting piece **27**′.
(46) Consequently, the screws **32** and **40** do not interact with an intermediate piece **30** but directly with the mounting piece **27**′. To this end, a threaded hole **36**′ for receiving the fastening screw **32** and a slot **50**′ for receiving the adjustment screw **40** are provided on a top surface of the mounting piece **27**′. Adjustment of the furniture hinge **2** by means of the screws **32**, **40** is similar to what was previously described in respect to the first embodiment.
(47) The cylinder **24** of the damping element **22** is fastened to one of the hinge parts **10**, **12** of the furniture hinge **2** (e.g. the hinge arm **12**) in a manner pivotable about a pivot axis **86** as described above in respect to the first embodiment. Similarly, a distal end of the piston rod **26** of the damping element **22** is attached to the other component (e.g. the first hinge lever **18**.**1** or its second lever arm, respectively) pivotably about the articulation axis **90**.
(48) The dome-shaped mounting piece **27**′ forms a hollow space **108** between its top surface and

the mounting surface of the furniture body **8**, the hollow space **108** extending parallel to the longitudinal direction **38**. The hollow space **108** serves for receiving the cylinder **24** of the damping element **22** during its pivot movement about pivot axis **86** during an opening movement and/or a closing movement of the furniture hinge **2**. This allows the design of a particular compact furniture hinge **2**.

Claims

1. A furniture hinge for fastening a furniture door or flap of a piece of furniture to a furniture body of the piece of furniture in an articulated manner, the furniture hinge comprising: a first hinge part configured to be attached to the furniture door or flap; a mounting piece configured to be attached to the furniture body; a second hinge part configured to be attached to the mounting piece; a hinge assembly connecting the first hinge part to the second hinge part to permit an opening movement of the first hinge part away from the second hinge part and an oppositely directed closing movement; and at least one dampening element configured to dampen the opening movement and/or the closing movement at least towards an end of the movement, the dampening element including a cylinder, a piston linearly movable in the cylinder, and a piston rod coupled to the piston and projecting from the cylinder, the cylinder being pivotably connected about a pivot axis directly or indirectly to one of the first and second hinge parts at least during the opening movement and/or the closing movement; wherein the at least one dampening element further includes a bearing element arranged on an outer circumferential surface of the cylinder between ends of the cylinder, wherein the cylinder is pivotably connected about the pivot axis to the one of the first and second hinge parts by the bearing element.
2. The furniture hinge of claim 1, wherein: the other of the first and second hinge parts to which the cylinder is not pivotably connected is connected directly or indirectly to the piston rod at least towards an end of the opening movement and/or the closing movement.
3. The furniture hinge of claim 1, wherein: the hinge assembly includes at least one hinge lever connected to the first hinge part pivotable about a first hinge axis and connected to the second hinge part pivotable about a second hinge axis, the piston rod being connected to the at least one hinge lever at least towards an end of the opening movement and/or the closing movement, so that the at least one hinge lever actuates the dampening element.
4. The furniture hinge of claim 1, wherein: the hinge assembly includes at least a first hinge lever and a second hinge lever, the first hinge lever being connected to the first hinge part pivotable about a first hinge axis and connected to the second hinge part pivotable about a second hinge axis, the second hinge lever being connected to the first hinge part pivotable about a third hinge axis and connected to the second hinge part pivotable about a fourth hinge axis, the piston rod being connected to the first hinge lever at least towards an end of the opening movement and/or the closing movement so that the first hinge lever actuates the dampening element.
5. The furniture hinge of claim 1, wherein: the other of the first and second hinge parts to which the cylinder is not pivotably connected is connected directly or indirectly to the piston rod so that the piston rod is pivotable about an articulation axis relative to the other of the first and second hinge parts.
6. The furniture hinge of claim 1, wherein: the hinge assembly includes at least a first hinge lever and a second hinge lever, the first hinge lever being connected to the first hinge part pivotable about a first hinge axis and connected to the second hinge part pivotable about a second hinge axis, the second hinge lever being connected to the first hinge part pivotable about a third hinge axis and connected to the second hinge part pivotable about a fourth hinge axis, the piston rod being pivotably directly connected about an articulation axis to the first hinge lever, wherein when the furniture hinge is mounted on the piece of furniture a surface of the furniture body to which the mounting piece is attached defines a first plane, wherein a second plane is defined which extends

parallel to the first plane and passes through the second hinge axis, wherein a third plane is defined which extends parallel to the first plane and passes through the fourth hinge axis, and wherein the articulation axis is located between the second plane and the third plane throughout the entire opening movement and/or closing movement of the furniture hinge.
7. The furniture hinge of claim 6, wherein: a distance between the articulation axis and a straight line connecting the second hinge axis with the fourth hinge axis increases continuously during the entire opening movement of the furniture hinge.
8. The furniture hinge of claim 1, wherein: the second hinge part is movably connected to the mounting piece and the second hinge part can be fixed to the mounting piece in a selected position relative to the mounting piece, such that by changing the selected position of the second hinge part relative to the mounting piece the furniture door or flap can be adjusted in at least one direction relative to the furniture body.
9. The furniture hinge of claim 1, wherein: the second hinge part is releasably connected to the mounting piece such that by releasing the second hinge part from the mounting piece the furniture door or flap is released from the furniture body.
10. The furniture hinge of claim 1, further comprising: at least one spring connected directly or indirectly between the first hinge part and the second hinge part and supporting the opening movement and/or the closing movement at least towards the end of the movement, the at least one spring urging the furniture door or flap into an open position and/or a closed position with respect to the furniture body.
11. The furniture hinge of claim 10, wherein: the at least one spring is a torsion spring.
12. The furniture hinge of claim 10, wherein: the at least one spring is supported at a first end directly or indirectly from the first hinge part and the at least one spring is supported at a second end directly or indirectly from the second hinge part.
13. The furniture hinge of claim 1, in combination with the piece of furniture such that the furniture door or flap is fastened to the furniture body in an articulated manner by the furniture hinge.
14. The furniture hinge of claim 1, wherein: the cylinder of the dampening element is pivotably connected to the one of the first and second hinge parts such that a pivot position of the cylinder with respect to the one of the first and second hinge parts changes during the opening movement and/or the closing movement.
15. A furniture hinge for fastening a furniture door or flap of a piece of furniture to a furniture body of the piece of furniture in an articulated manner, the furniture hinge comprising: a first hinge part configured to be attached to the furniture door or flap; a mounting piece configured to be attached to the furniture body; a second hinge part configured to be attached to the mounting piece; a hinge assembly connecting the first hinge part to the second hinge part to permit an opening movement of the first hinge part away from the second hinge part and an oppositely directed closing movement; and at least one dampening element configured to dampen the opening movement and/or the closing movement at least towards an end of the movement, the dampening element including a cylinder, a piston linearly movable in the cylinder, and a piston rod coupled to the piston and projecting from the cylinder, the cylinder being pivotably connected about a pivot axis directly or indirectly to one of the first and second hinge parts at least during the opening movement and/or the closing movement; wherein the hinge assembly includes at least a first hinge lever and a second hinge lever, the first hinge lever being connected to the first hinge part pivotable about a first hinge axis and connected to the second hinge part pivotable about a second hinge axis, the second hinge lever being connected to the first hinge part pivotable about a third hinge axis and connected to the second hinge part pivotable about a fourth hinge axis, the piston rod being pivotably connected about an articulation axis to the first hinge lever, wherein when the furniture hinge is mounted on the piece of furniture a surface of the furniture body to which the mounting piece is attached defines a first plane, wherein a second plane is defined which extends parallel to the first plane and passes through the second hinge axis, wherein a third plane is defined which extends parallel to the

first plane and passes through the fourth hinge axis, and wherein the articulation axis is located between the second plane and the third plane throughout the entire opening movement and/or closing movement of the furniture hinge; and wherein a distance between the articulation axis and a straight line connecting the second hinge axis with the fourth hinge axis increases continuously during the entire opening movement of the furniture hinge.

16. A furniture hinge for fastening a furniture door or flap of a piece of furniture to a furniture body of the piece of furniture in an articulated manner, the furniture hinge comprising: a first hinge part configured to be attached to the furniture door or flap; a mounting piece configured to be attached to the furniture body; a second hinge part configured to be attached to the mounting piece; a hinge assembly connecting the first hinge part to the second hinge part to permit an opening movement of the first hinge part away from the second hinge part and an oppositely directed closing movement; and at least one dampening element configured to dampen the opening movement and/or the closing movement at least towards an end of the movement, the dampening element including a cylinder, a piston linearly movable in the cylinder, and a piston rod coupled to the piston and projecting from the cylinder, the cylinder being pivotably connected about a pivot axis directly or indirectly to one of the first and second hinge parts at least during the opening movement and/or the closing movement; wherein the at least one dampening element further includes a bearing element arranged on an outer circumferential surface of the cylinder between ends of the cylinder, wherein the cylinder is pivotably connected about the pivot axis to the one of the first and second hinge parts by the bearing element; and wherein the pivot axis of the pivotable connection of the cylinder to the one of the first and second hinge parts extends at a distance greater than zero from a longitudinal axis of the cylinder.